Selected boundary stability for periodic mean-field games

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Abstract

For every dimension, smooth local coupling and classical genuinely periodic quadratic mean-field-game equilibrium, we characterize a specified weaker nonlinear stability property: all small smooth boundary perturbations admit a nearby selected equilibrium with an error linear in the boundary error, with constants depending on the horizon. The criterion is both necessary and sufficient, including a negative conclusion at every conjugate horizon. It has finite matrix certificates constructed from two scalar parabolic problems, and a convergent equilibrium-selection rule. An explicit entire periodic family has only a discrete set of conjugate horizons.

Scope. This addresses the weaker-stability alternative in C73-40 for its entire original model class. No restriction to a special coupling, dimension, periodic family or a preselected set of regular horizons is made in the general theorem. The original horizon-uniform orbital shadowing of every admissible equilibrium is a stronger question and is not claimed here.

1 The original boundary-value setting

Let $\nu > 0$, let $f : \mathbb{T}^d \times (0,\infty) \to \mathbb{R}$ be smooth, and let a positive, mass-one classical reference satisfy

$$\begin{aligned} \lambda_* - v_{*,t} - \nu\Delta v_* + \tfrac{1}{2}|Dv_*|^2 &= f(x, m_*), \\ m_{*,t} - \nu\Delta m_* - \operatorname{div}(m_* Dv_*) &= 0, \\ (v_*, m_*)(t+P) = (v_*, m_*)(t), \qquad & \int m_*(t) = 1. \end{aligned} \tag{MFG}$$

The pair (Dv_*, m_*) is nonconstant in time. The actual reference value $v_*(t,x) - \lambda_* t$ may drift. Its orbit is $\mathcal{O}_* = \{(Dv_*(s), m_*(s)) : s \in \mathbb{R}/P\mathbb{Z}\}$, and its distance is the infimum over s of the sum of the gradient L^∞ error and density L^1 error.

Perturbed equilibria solve

$$-u_t - \nu\Delta u + \tfrac{1}{2}|Du|^2 = f(x,m), \qquad m_t - \nu\Delta m - \operatorname{div}(mDu) = 0, \qquad m(0) = m_0, \quad u(T) = h.$$

Initial population and terminal value are the independent boundary data. No initial-value MFG semigroup is presumed.

Cardaliaguet–Delarue discuss the periodic-solution stability question in [1, Section 3.3, paragraph "Periodic solutions," p. 3681]. Genuine periodic examples already appear in [2, Section 6, Theorem 6.1 and Remark 6.4]. The classification proved here concerns the specified nonlinear selection property, rather than imaginary eigenvalues.

2 The nonlinear property being classified

Write $Q_T = [0,T] \times \mathbb{T}^d$, with the torus of volume one, and $\Pi z = z - \int_{\mathbb{T}^d} z$. Fix $0 < \alpha < 1$. For any reference in the original problem, any $T > 0$, and any phase θ, put

$$\bar m(t) = m_*(t+\theta), \quad b(t) = Dv_*(t+\theta), \quad a(t,x) = f_m(x, \bar m(t,x)), \quad u_{\mathrm{r}}(t,x) = v_*(t+\theta,x) - \lambda_* t.$$

Both $\bar{m}$ and b are known reference fields. We use

$$X_T = \{(w,\rho) \in (C^{1+\alpha/2,2+\alpha}(Q_T))^2 : \int w(t) = \int \rho(t) = 0\}, \qquad D = (C_0^{2+\alpha}(\mathbb{T}^d))^2,$$

with their usual sum norms; a subscript zero means spatial mean zero. The boundary perturbation is $d = (d_0, d_T) = (m_0 - \bar{m}(0), \Pi(h - u_{\mathrm{r}}(T)))$. Every sufficiently small $d \in D$ gives a positive normalized initial density. Terminal constants are free and will be restored explicitly.

Selected local Lipschitz boundary stability at (T,θ). There are $r, \delta, C > 0$ and a rule $S : \{\|d\|_D < \delta\} \to X_T$ such that $S(0) = 0$, and for every such d, $S(d) = (w,\rho)$ produces a classical MFG equilibrium with the stated initial density and terminal value, with

$$\|(w,\rho)\|_{X_T} \le C\|d\|_D < r, \qquad \bar{m} + \rho \ge \tfrac{1}{2}\min_{Q_T} \bar{m}. \tag{W1}$$

No continuity or differentiability of S is assumed in this definition. The theorem constructs a Lipschitz rule when the property holds. This is a specified weaker alternative in C73-40: boundary errors are measured more strongly, a nearby equilibrium is selected, and the constants may depend on horizon and phase. In particular,

$$\sup_{0\le t\le T} d_*((Du(t), m(t)), \mathcal{O}_*) \le \|(w,\rho)\|_{X_T} \le C\|d\|_D. \tag{W2}$$

The comparison phase in this estimate is $t + \theta$ modulo P. The neighborhood can always be chosen with density lower bound $\min m_*/2$ and an admissibility upper bound one larger than the reference norm. If the original specified caps have strict slack, it can also be chosen within those caps. We do not assert an open perturbation result inside an arbitrarily saturated cap.

For a set $H \subset (0,\infty) \times (\mathbb{R}/P\mathbb{Z})$, the reference has this property on H when it has it at every pair in H. The criterion below classifies every pair, including the pairs where the property fails.

3 Scalar parabolic facts and an explicit compactness bound

We record the analytic facts used in the classification. All coefficients below are smooth on a compact cylinder and $\nu > 0$.

Lemma 1 (Scalar equations)**.** *The forward equation*

$$y_t - \nu\Delta y + c\cdot Dy + ey = g, \qquad y(0) = 0,$$

and its time reversal have a unique solution in $\mathcal{W}_T = H^1(0,T;L^2) \cap L^2(0,T;H^2)$ for each $g \in L^2$. For parabolic $C^{\alpha/2,\alpha}$ forcing and $C^{2+\alpha}$ initial or terminal data the solution is in $C^{1+\alpha/2,2+\alpha}$, with a bounded solution operator. Writing $B = \|c\|_\infty$ and $E = \|e\|_\infty$, define

$$\begin{aligned} A &= 1 + 2E + B^2/\nu, & Y &= \sqrt{T}e^{AT/2}, \\ G &= \sqrt{(1+AY^2)/\nu}, & J &= \frac{\sqrt{3}}{\nu}\sqrt{1 + B^2G^2 + E^2Y^2}, \\ R(T,\nu,B,E) &= Y + G + J + 1 + \nu J + BG + EY. \end{aligned} \tag{P1}$$

Then $\|y\|_2 \le Y\|g\|_2$, $\|Dy\|_2 \le G\|g\|_2$, $\|\Delta y\|_2 \le J\|g\|_2$, and $\|y_t\|_2 \le (1 + \nu J + BG + EY)\|g\|_2$. All these norms are on Q_T.

Proof. Multiplication by y, Young's inequality, and integration give

$$(\|y(t)\|_2^2)' + \nu\|Dy(t)\|_2^2 \le \|g(t)\|_2^2 + A\|y(t)\|_2^2.$$

Gronwall gives the bounds Y, G. Multiplication by $-\Delta y$ gives

$$(\|Dy(t)\|_2^2)' + \nu\|\Delta y(t)\|_2^2 \le \frac{3}{\nu}(\|g(t)\|_2^2 + B^2\|Dy(t)\|_2^2 + E^2\|y(t)\|_2^2).$$

This proves the bound J, and the equation proves the time derivative bound. Spatial Fourier truncations give finite linear ODEs. These uniform estimates yield a weak limit with the stated zero trace; the first estimate applied to a difference proves uniqueness. On a torus, the L^2 norm of the full spatial Hessian equals that of Δy, so the asserted space is exactly controlled.

The Hölder assertion follows directly from the heat kernel. If $H_t = e^{\nu t\Delta}$ and $z(t) = \int_0^t H_{t-s}q(s)\,ds$, the periodic heat kernel satisfies

$$\int |D^j H_t(x)|\,\operatorname{dist}(x,0)^\alpha\,dx \le Ct^{(\alpha-j)/2} \quad (0<t\le T),$$

with the analogous bound for a time derivative (which adds two to j). These follow by differentiating the Gaussian and summing its translates. The integral of every second spatial derivative is zero. Subtracting $q(s,x)$ inside the second-derivative convolution therefore gives the integrable majorant $C\|q\|_\alpha(t-s)^{\alpha/2-1}$. For a spatial increment h, split the integral at h^2: the near part is bounded by $C|h|^\alpha$, and the far part by $C|h|\int_{h^2}^T r^{\alpha/2-3/2}dr \le C|h|^\alpha$. For a time increment $h>0$, splitting at h and using a heat-kernel time derivative bounds the two parts by $Ch^{\alpha/2}$ and $Ch\int_h^T r^{\alpha/2-2}dr \le Ch^{\alpha/2}$. Thus D^2z has the required parabolic Hölder bound, and $z_t = q+\nu\Delta z$ does too. The homogeneous heat solution with $C^{2+\alpha}$ data satisfies the same estimates by commuting spatial derivatives and the same kernel bounds.

To include $c\cdot Dy + ey$, use the elementary interpolation estimate

$$\|Dy\|_{C^{\alpha/2,\alpha}} + \|y\|_{C^{\alpha/2,\alpha}} \le \varepsilon\|y\|_{C^{1+\alpha/2,2+\alpha}} + C_\varepsilon\|y\|_\infty.$$

It follows by finite spatial differences and splitting increments into those smaller and larger than a fixed interpolation length; time increments use the controlled time derivative. Product Hölder inequalities and a sufficiently small ε absorb the highest norm. The maximum principle bounds the remaining supremum norm by $e^{ET}(\|y(0)\|_\infty + T\|g\|_\infty)$. Consequently the Hölder estimate is uniform for the coefficients sc, se, $0\le s\le 1$. The heat equation is solvable at $s=0$. The bounded inverse estimate makes solvability open by the Neumann series. It makes solvability closed by compactness in every lower Hölder exponent and the same estimate for the limiting equation. This proves existence at $s=1$; the energy estimate proves uniqueness. Time reversal proves the terminal version. Projecting a scalar backward solution onto its spatial mean-zero part proves the same assertion when its equation is spatially centered. □

Classical entire periodic reference solutions in the original statement are smooth, so the lemma does not strengthen that statement's reference assumptions. Indeed m_t, D^2m, v_t, D^2v are bounded on the periodic cylinder. Spatial interpolation shows Dv is $1/2$-Hölder in time and Lipschitz in space, while m is Lipschitz in time and space. The HJB right-hand side $f(x,m)-|Dv|^2/2$ is consequently parabolic $C^{\alpha/2,\alpha}$. Interior versions of the heat-kernel estimates just proved upgrade v. In nondivergence form the population equation then has Hölder right-hand side $Dv\cdot Dm + m\Delta v$, upgrading m. Differentiating and repeating on interior subcylinders upgrades both fields to every order. Periodicity covers the entire cylinder without an initial or terminal regularity exception.

4 A criterion using scalar problems and finite matrices

Let $\mathcal{H}_T = L^2(Q_T)$ with spatial mean zero at almost every time. For $r\in\mathcal{H}_T$, first solve the centered scalar terminal problem and then the scalar initial problem

$$\begin{aligned} -W_t - \nu\Delta W + \Pi(b\cdot DW) &= \Pi(ar), & W(T)&=0,\\ Z_t - \nu\Delta Z - \operatorname{div}(bZ) &= \operatorname{div}(\bar m DW), & Z(0)&=0. \end{aligned} \tag{K1}$$

Define $K_{T,\theta}r = Z$. These are two uniquely solvable scalar equations. Put $B=\|b\|_\infty$, $E=\|\operatorname{div} b\|_\infty$, $A_0=\|a\|_\infty$, $M=\|\bar m\|_\infty$, $M_1=\|D\bar m\|_\infty$. Let G_0, J_0 be (P1) with $E=0$ and set

$$C_K = 1 + R(T,\nu,B,E)A_0(MJ_0 + M_1G_0). \tag{K2}$$

Lemma 1 proves

$$\|Kr\|_2 + \|(Kr)_t\|_2 + \|\Delta Kr\|_2 \le C_K \|r\|_2.$$

Thus K is compact: bounded sets in $\mathcal{W}_T$ are compact in L^2, as is also demonstrated by the explicit projections below.

Let P_N be orthogonal projection onto temporal cosine modes $0 \le n \le N$ and spatial Fourier modes $0 < \|k\|_\infty \le N$. Use real sine–cosine combinations for a real orthonormal basis $e_1, \dots, e_{q_N}$ of its range, where $q_N = (N+1)((2N+1)^d - 1)$. Parseval and integration by parts against the temporal sine modes give

$$\|(I - P_N)K\| \le e_N := \frac{C_K(T+1)}{N+1}. \tag{K3}$$

More precisely, the time tail is bounded by $T\|(Kr)_t\|_2/(\pi(N+1))$ and the spatial tail by $\|\Delta Kr\|_2/(4\pi^2(N+1)^2)$; their sum implies (K3). Define the finite real matrix

$$D_N = (\delta_{ij} - \langle e_i, Ke_j \rangle)_{i,j=1}^{q_N}. \tag{K4}$$

Every entry is specified by (K1), using only the original coefficients. The following is an exact finite-certificate criterion:

$$\boxed{\exists N \ge 1: \quad \det D_N \ne 0, \qquad e_N\{1 + (1 + C_K)\|D_N^{-1}\|_{2\to 2}\} < 1.} \tag{FC}$$

No precision or running-time assertion is implicit: arbitrary smooth coefficients need not even be computably presented. Certified bounds on the scalar problems and finite matrix entries can verify a strict certificate when those data are effectively supplied.

Lemma 2 (Exactness of the finite certificate)**.** *Condition (FC) holds if and only if $I - K$ is invertible on $\mathcal{H}_T$.*

Proof. In $\operatorname{ran} P_N \oplus \ker P_N$, the operator $I - P_N K$ is upper triangular, with diagonal blocks D_N, I and upper right block $-P_N K(I - P_N)$. If D_N is invertible, its full inverse has norm at most

$$B_N = 1 + (1 + C_K)\|D_N^{-1}\|.$$

Since $\|(I - K) - (I - P_N K)\| \le e_N$, the Neumann series proves invertibility whenever $e_N B_N < 1$. Conversely, if $I - K$ is invertible, (K3) implies that, for all sufficiently large N, $I - P_N K$ is invertible with inverse norm at most $2\|(I-K)^{-1}\|$. Restriction to its finite invariant subspace bounds $\|D_N^{-1}\|$ by the same constant. Thus B_N is bounded and $e_N B_N < 1$ eventually. This proves both directions. □

5 Fredholm reduction of the Jacobi boundary problem

The centered linear boundary operator is

$$L(w, \rho) = \big(\Pi[-w_t - \nu\Delta w + b \cdot Dw - a\rho],\ \rho_t - \nu\Delta\rho - \operatorname{div}(b\rho + \bar{m}Dw),\ \rho(0), w(T)\big). \tag{L1}$$

Its target is the product of two mean-zero parabolic $C^{\alpha/2,\alpha}$ spaces and D. In particular it includes both interior residuals as well as the two boundary data.

Lemma 3 (General Fredholm reduction)**.** *For every original model, phase and finite horizon, L has Fredholm index zero. It is invertible if and only if its homogeneous boundary problem has no nonzero solution, and this is also equivalent to (FC).*

Proof. Delete $-a\rho$ from the first component of L, obtaining L_0. The scalar lemma solves first the terminal value equation for w and then the initial population equation for ρ; hence L_0 is an isomorphism on the indicated Hölder spaces. The deleted term factors through the compact embedding $C^{1+\alpha/2,2+\alpha} \hookrightarrow C^{\alpha/2,\alpha}$. Thus L is an isomorphism plus a compact operator and has index zero. The compact-operator alternative gives its invertibility precisely when its kernel is zero. One elementary

justification of that alternative is to approximate the compact operator in norm by a finite-rank operator with remainder less than one, invert the remainder by a Neumann series, and reduce kernel and cokernel to the same finite square matrix. The compactness here permits finite-rank approximation by spatial–temporal interpolation, so no Banach approximation property is presumed.

For a zero-boundary solution, the scalar reduction (K1) says exactly $\rho = K\rho$, and conversely such a fixed density reconstructs w. This is true in L^2 and in the Hölder spaces. These kernels agree. To see this without a dimension restriction, an L^2 kernel first gives $w, \rho \in \mathcal{W}_T$. All their spatial derivatives of order at most two are then in L^2. Suppose derivatives through order j are known in L^2. Differentiate the backward scalar equation to order j. Its leading forcing is $aD^j\rho$; commutators contain only already controlled derivatives. The zero terminal trace remains zero, so the scalar lemma puts $D^j w$ in $\mathcal{W}_T$. Next differentiate the forward equation: its principal forcing is $\operatorname{div}(\bar{m}D(D^j w))$; commutators contain derivatives of ρ through order j and of w through order $j+1$, now controlled. Its zero initial trace again gives $D^j\rho \in \mathcal{W}_T$. Spatial difference quotients justify these distributional differentiations and the zero traces. This gains two orders and iterates indefinitely. The elementary trace estimate $H^1_t H^s_x \cap L^2_t H^{s+2}_x \subset C_t H^{s+1}_x$ (obtained by differentiating the squared H^{s+1} Fourier norm) then gives continuity of every spatial derivative. The equations give all time derivatives in succession. Both fields are smooth up to their respective time endpoints and belong to the Hölder kernel. Lemma 2 completes the equivalence. □

6 Boundary traces detect every nonzero Jacobi kernel

Lemma 4 (Zero at both endpoints)**.** *A homogeneous centered Jacobi field (w, ρ) for which $w(0) = \rho(0) = w(T) = \rho(T) = 0$ is identically zero.*

Proof. Set $\kappa = \bar{m}/(2\nu)$, $r = \rho + \kappa w$, $U = (w, r)$ and $A = \operatorname{diag}(\nu(-\Delta), -\nu(-\Delta))$ on the real L^2 product space. The first component has spatial mean zero; the second need not. Let

$$c_w(t) = \int (b \cdot Dw - a\rho) = \int [-(\operatorname{div} b)w - ar + a\kappa w].$$

Using $\bar{m}_t = \nu\Delta\bar{m} + \operatorname{div}(\bar{m}b)$, direct substitution cancels all cross second derivatives and gives

$$\begin{aligned} w_t &= \nu(-\Delta)w + b \cdot Dw + a\kappa w - ar - c_w, \\ r_t &= -\nu(-\Delta)r + b \cdot Dr + (\operatorname{div} b - a\kappa)r + a\kappa^2 w - \kappa c_w. \end{aligned}$$

Thus $U_t = AU + B(t)U$, where A is a constant self-adjoint operator and $B(t)$ has only first spatial derivatives and bounded terms. Integration by parts in the expression for c_w makes its bounded rank-one character explicit. For some finite C_0,

$$\|B(t)U\|^2 \le C_0(\langle |A|U, U\rangle + \|U\|^2).$$

Both endpoint traces of U vanish. Put $V = e^{\phi(t)}U$, where $\phi' > 0$ will be chosen. Integration by parts yields the exact identity

$$\int_0^T \|V_t - (A + \phi')V\|^2 dt = \int_0^T \{\|V_t\|^2 + \|(A + \phi')V\|^2 + \phi''\|V\|^2\} dt.$$

The endpoint terms are zero. Spatial Fourier truncation followed by the $\mathcal{W}_T$ limit justifies the identity at this regularity. On the other hand $V_t - (A + \phi')V = BV$, and

$$\langle |A|V, V\rangle \le \|AV\|\|V\| \le \|(A + \phi')V\|\|V\| + \phi'\|V\|^2.$$

Young's inequality therefore bounds the left side by

$$\tfrac{1}{2}\int_0^T \|(A + \phi')V\|^2 dt + C_1 \int_0^T (1 + \phi')\|V\|^2 dt.$$

Choose $\phi'(t) = qe^{kt}$ with $q \ge 1$ and $k > 2C_1$. Then $\phi'' = k\phi' > C_1(1 + \phi')$, so the two displays force $V = 0$. This proves the lemma. The proof uses no forward MFG semigroup or sign assumption on a. □

For a zero-boundary Jacobi field (z,σ), namely $\sigma(0)=0, z(T)=0$, and a perturbation (w,ρ) with interior linear residuals (F,G), integration by parts gives

$$\int d_T\sigma(T)+\int z(0)d_0=-\int_{Q_T}(F\sigma+zG). \tag{G1}$$

Indeed differentiating $\int(w\sigma-z\rho)$ cancels both Laplacians, both drift terms, and the symmetric coupling terms, leaving $-\int(F\sigma+zG)$. Mean projections disappear because z,σ have mean zero. If both boundary traces $z(0),\sigma(T)$ are zero, Lemma 4 says the field is zero. Consequently every nonzero kernel field gives a nonzero functional on the independent boundary directions $(d_0,d_T)\in D$.

7 Exact nonlinear classification and the selection rule

Theorem 5. *For every reference allowed in C73-40, every $0<\alpha<1$, every $T>0$, and every phase θ, the following are equivalent:*

1. *Selected local Lipschitz boundary stability (W1) holds.*
2. *The homogeneous Jacobi problem (L1) has zero kernel.*
3. *The finite-certificate condition (FC) holds.*

When they hold, the rule below selects the unique nearby equilibrium and is Lipschitz between boundary data; it also satisfies (W2). When they fail, no rule satisfying (W1) for all sufficiently small independent boundary perturbations exists, even without continuity of that rule. A reference has the property on a specified set H exactly when (FC) holds at every pair in H.

Proof. The equivalence of items 2 and 3 is Lemma 3. Write the nonlinear perturbation equations as

$$L\xi+N(\xi)=(0,0,d_0,d_T),\qquad \xi=(w,\rho),$$

where

$$N(\xi)=\Big(\Pi\left[\tfrac12|Dw|^2-\{f(x,\bar m+\rho)-f(x,\bar m)-a\rho\}\right],-\operatorname{div}(\rho Dw),0,0\Big). \tag{N1}$$

On a sufficiently small positive-density ball in X_T, Taylor's integral formula and the product Hölder inequality give

$$\|N(\xi)\|\le C_2\|\xi\|^2,\qquad \|N(\xi)-N(\eta)\|\le C_2(\|\xi\|+\|\eta\|)\|\xi-\eta\|. \tag{N2}$$

Explicitly, the scalar Taylor remainder is $\rho^2\int_0^1(1-s)f_{mm}(x,\bar m+s\rho)ds$; its multiplier and its Lipschitz dependence have bounded Hölder norms on the fixed compact positive density interval. The other components are products of the derivatives controlled by X_T.

If item 2 holds, let $K_L=\|L^{-1}\|$. Choose $r>0$ below the positive-density and declared admissibility margins and with $2C_2K_Lr\le 1/2$. Take $\delta\le r/(2K_L)$. Starting at $\xi_0=0$, perform the explicit iteration

$$\xi_{n+1}=L^{-1}\{(0,0,d_0,d_T)-N(\xi_n)\}. \tag{SEL}$$

It maps the closed ball of radius r into itself, since its norm is at most $K_L\|d\|+K_LC_2r^2\le 3r/4$, and its contraction factor is at most $2K_LC_2r\le 1/2$. Successive differences sum geometrically, proving convergence and uniqueness in this ball. Its limit satisfies

$$\|\xi(d)\|\le K_L\|d\|+\tfrac12\|\xi(d)\|,\qquad \|\xi(d)-\xi(d')\|\le 2K_L\|d-d'\|.$$

The first inequality gives (W1), and the second gives the claimed Lipschitz selection. To implement each linear step, the scalar reduction solves $I-K$; a certificate (FC) gives the convergent Neumann series around $I-P_NK$ from Lemma 2.

The projected first equation has a spatially constant residual $r_u(t)$ for $u_{\mathrm r}+w$. Choose a scalar $c(t)$ with $c'(t)=r_u(t)$ and $c(T)=\int(h-u_{\mathrm r}(T))$. Then $u=u_{\mathrm r}+w+c$ solves the unprojected HJB equation

and has exactly terminal value h. The population equation, initial condition, mass one, and positivity already hold. This proves actual nonemptiness for every boundary datum, and (W2) follows by the reference phase $t+\theta$.

Conversely suppose item 1 holds and a nonzero zero-boundary Jacobi field (z,σ) exists. For any fixed boundary direction $d \in D$, choose $\varepsilon > 0$ small enough that εd lies in the open permitted boundary neighborhood. Its selected perturbation has norm at most $C\varepsilon\|d\|$ and interior linear residuals $-(N_1, N_2)$. Equation (G1) yields

$$\varepsilon\left(\int d_T\sigma(T) + \int z(0)d_0\right) = \int_{Q_T}(N_1\sigma + zN_2) = O(\varepsilon^2).$$

Divide by ε and let it decrease to zero. Since the boundary direction was arbitrary, both mean-zero traces $z(0), \sigma(T)$ vanish. Lemma 4 contradicts nonzeroness. This proves necessity, including the negative classification at every conjugate pair. It rules out exactly (W1), not every continuous selector or the original orbital property. □

On a compact set of horizon–phase pairs where the criterion holds, the above constants and the radius may be chosen uniformly. Rescale time to $[0,1]$; the coefficients and L vary continuously in operator norm. The inverse Neumann estimate is local in those parameters, so compactness gives a uniform inverse bound. Reference positivity and the product estimates are also uniform there. This argument supplies no uniform bound on an unbounded set of horizons or across a singular pair.

8 An entire periodic family: every horizon is classified

The general criterion applies to explicit genuine periodic equilibria. Work on $\mathbb{T}^1$ and fix $0 < r < 1$, $c \neq 0$, $\nu > 0$. Put $k = 2\pi$, $s = \sqrt{1-r^2}$ and

$$\begin{aligned} M(y) &= 1 + r\cos(ky), \\ p(y) &= c\left(\frac{s}{M(y)} - 1\right) - \nu\frac{M'(y)}{M(y)}, \\ f(z) &= \frac{s^2(c^2+\nu^2k^2)}{2z^2}, \qquad \lambda = \frac{c^2+\nu^2k^2}{2}. \end{aligned} \tag{TW1}$$

Since $\int M^{-1} = s^{-1}$ and $\int M'/M = 0$, p has mean zero. Let $V' = p$ with $\int V = 0$. The fields

$$v_*(t,x) = V(x-ct), \qquad m_*(t,x) = M(x-ct) \tag{TW2}$$

solve the original periodic MFG system, with period $P = 1/|c|$. Indeed, setting $b_0 = p + c$ gives $b_0 M = cs - \nu M'$, so $-\nu M'' - (b_0 M)' = 0$. Furthermore $M'' = -k^2(M-1)$ and $(M')^2 = k^2(r^2 - (M-1)^2)$ give

$$\tfrac{1}{2}b_0^2 - \nu b_0' = \frac{s^2(c^2+\nu^2k^2)}{2M^2} - \frac{\nu^2k^2}{2}.$$

Using $cp + p^2/2 = (b_0^2 - c^2)/2$ proves $\lambda + cp - \nu p' + p^2/2 = f(M)$, the required HJB equation. The density is positive, has mass one, and is genuinely time-dependent. The integral formula for M^{-1} follows, for example, by the substitution $z = \tan(ky/2)$ on the two half-circles.

Theorem 6 (All horizons for the explicit family)**.** *For every triple (r,c,ν) above there is a set $\Sigma \subset (0,\infty)$, bounded away from zero and finite in every bounded interval, such that at every phase:*

- *For $T \notin \Sigma$, selected local Lipschitz stability (W1) holds, with the convergent selection (SEL).*
- *For $T \in \Sigma$, no selection satisfying (W1) for all small independent boundary perturbations exists.*

The set is exactly the zero-kernel failure set in Theorem 5, so (FC) classifies its complement without an assumed mixed-boundary inverse. The constants can be uniform over phase and any compact subset of $(0,\infty)\setminus\Sigma$.

Proof. Use $y = x - c(t+\theta)$, $\tau = t/T$, and $q(y) = f'(M(y))$. The uncentered linear fields satisfy, on the fixed cylinder $[0,1]\times\mathbb{T}$,

$$\begin{aligned} B_T W &:= -W_\tau - T\nu W_{yy} + Tb_0 W_y = TqR, \qquad W(1) = 0, \\ F_T R &:= R_\tau - T\nu R_{yy} - T(b_0 R)_y = T(MW_y)_y, \quad R(0) = 0. \end{aligned} \tag{TW3}$$

Its density has mean zero. Allowing the scalar value mean here changes neither its gradient nor the density Schur operator from (K1); centering the scalar solution recovers exactly that operator.

Let $\mathcal{H} = L^2([0,1]\times\mathbb{T})$, $\mathcal{H}_0 = \{R : \int R(\tau) = 0\}$, and $\mathfrak{D} = H^1_\tau L^2_y \cap L^2_\tau H^2_y$. Use the fixed closed domain $W(1) = 0$ for B_T, and the fixed domain $R(0) = 0$, $R \in \mathcal{H}_0$ for F_T. The scalar lemma makes both maps bounded isomorphisms onto their respective L^2 targets for every real $T > 0$. Define

$$\mathcal{K}_T = T^2 F_T^{-1} \partial_y (M\partial_y) B_T^{-1} q : \mathcal{H}_0 \longrightarrow \mathcal{H}_0. \tag{TW4}$$

The derivative operator is bounded from $\mathfrak{D}$ to $\mathcal{H}_0$, and the outer forward inverse maps into $\mathfrak{D}$, whose embedding in $\mathcal{H}$ is compact by the same Fourier–cosine tail estimate as (K3). Thus $\mathcal{K}_T$ is compact, including both time endpoints.

On these common domains the scalar operators are affine in T. Around any real $T_0 > 0$, their inverses are therefore holomorphic for sufficiently small complex $|T - T_0|$, by the inverse Neumann series. The union Ω of such disks is connected: it contains the entire positive real axis and every disk meets that axis. The inverse formulas agree on overlaps. Hence $\mathcal{K}_T$ is a holomorphic compact family on Ω.

There is an invertible point, rather than an identically singular family. For $0 < T \le 1$, the backward energy estimate for the first equation of (TW3) gives

$$\|W(\tau)\|_2 \le CT \int_\tau^1 \|R(s)\|_2 ds, \quad \|W\|_{L^2} \le CT\|R\|_{L^2}, \quad \|W_y\|_{L^2} \le C\sqrt{T}\|R\|_{L^2}.$$

Here and below C depends on the fixed coefficients, not on these small horizons. The first estimate follows by Gronwall applied to the spatial L^2 norm; testing the equation with W then gives the gradient estimate. For $Z = \mathcal{K}_T R$, the forward equation, tested with Z, gives

$$(\|Z\|_2^2)' + T\nu\|Z_y\|_2^2 \le CT\|Z\|_2^2 + CT\|W_y\|_2^2.$$

Gronwall yields $\|Z\|_{L^2} \le C\sqrt{T}\|W_y\|_{L^2} \le CT\|R\|_{L^2}$. Thus $\|\mathcal{K}_T\| \le CT < 1$ at all sufficiently small T. At any $z_0 \in \Omega$, choose a finite-rank orthogonal projection P with $\|(I-P)\mathcal{K}_{z_0}\| < 1/2$. With $Q = I - P$, this stays less than one near z_0, and the Q equation is solved by a Neumann series. The remaining finite matrix is

$$I - P\mathcal{K}_z P - P\mathcal{K}_z Q(I - Q\mathcal{K}_z Q)^{-1} Q\mathcal{K}_z P. \tag{TW5}$$

It is holomorphic, and singularity of $I - \mathcal{K}_z$ is exactly vanishing of its determinant. In a small connected disk either the zeros are isolated or the determinant vanishes identically. The set of points with an identically singular neighborhood is open and also closed: near a limit point use (TW5); its determinant vanishes on an open subdisk and hence throughout the disk by the scalar identity theorem. Connectedness of Ω and the small-T invertible point show that this open-and-closed set is empty. All singularities are therefore isolated in Ω. Their intersection Σ with the positive axis is locally finite; the small-horizon bound keeps it away from zero. The kernel bootstrap in Lemma 3 makes every L^2 kernel classical.

The moving coordinates give the same problem (TW3) at every phase. Changing phase changes only a constant spatial translation, an isometry of the boundary and perturbation norms. Finally Theorem 5 gives both the positive and negative nonlinear conclusions, and its compact-parameter argument proves the final uniformity assertion. □

9 What has been classified

For every original periodic reference, dimension, coupling, phase and finite horizon, Theorem 5 gives a necessary and sufficient criterion for (W1), including its failure when a Jacobi kernel is present. The

rule (SEL) gives actual equilibria for every small permitted boundary datum; positivity, mass and the terminal constant are included. Theorem 6 applies this general result to an explicit genuinely periodic family.

The constants are local to the stated horizon set and the boundary topology is $C^{2+\alpha}$. No asymptotic attraction, uniform-in-horizon all-equilibrium stability, finite decision algorithm or computational complexity bound is asserted.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

References

[1] P. Cardaliaguet and F. Delarue, *Selected topics in mean field games*, Proc. ICM 2022, vol. 5, pp. 3660–3703, EMS Press. DOI: 10.4171/ICM2022/17. Section 3.3, "Periodic solutions," printed p. 3681. https://ems.press/content/book-chapter-files/33265.

[2] M. Cirant and L. Nurbekyan, *The variational structure and time-periodic solutions for mean-field games systems*, Minimax Theory Appl. 3(2) (2018), 227–260. Theorem 6.1, arXiv v1 pp. 24–25; Remark 6.4, p. 28. https://arxiv.org/pdf/1804.08943v1.